SmartTicket: A Smart Event Ticketing System with Oracle PL/SQL

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GitHub: https://github.com/fumbaking/fumba_25401_smartTicket

1. Problem Statement

Rwanda's events sector is growing fast: MICE revenue reached US$95 million in 2023, 48% more than in 2022 (Kagina, 2024), and venues such as BK Arena sell concert tickets in tiers from 15,000 to 100,000 RWF (Iyamuremye, 2025). Ticketing has three weaknesses. **Scalping:** brokers buy underpriced seats early and resell them (Courty, 2003), and this rent-seeking wastes about a third of the efficiency gain that resale creates (Leslie & Sorensen, 2014). **Fraud:** a seat can be sold twice, or one QR code can be reused at the gate (Rafati Niya et al., 2022). **Opacity:** local platforms are closed, so organisers cannot see live occupancy, denied attempts or an audit trail. Promoters, fans and venue security are all affected.

2. Proposed Solution

SmartTicket is an Oracle pluggable database in which **all business rules are enforced in PL/SQL** and every seat follows one path: *one seat → one sale → one entry*. A booking succeeds only if the seat is free, the event is open, the buyer stays within 6 tickets per event, and the payment equals the booking total. Each ticket can be scanned in only once. Every action, including refused ones, is written to an audit log. **Main objective:** build an Oracle database that sells, validates and reports event tickets in real time while preventing double sales, scalping and fake entry. **Specific objectives:** (1) normalise a real events dataset to 3NF and load it; (2) build a booking, payment and cancellation package with exception handling; (3) enforce anti-fraud rules with triggers; (4) audit all changes and apply the weekday/holiday restriction rule; (5) deliver a live Power BI dashboard.

3. Data Source

SeatGeek Events & Ticket Listings Dataset (Rebrowser, 2026; Kaggle / Hugging Face; licence CC BY-NC 4.0, so educational use is allowed). It contains **1,364 events, 30,000 seat listings (section, row, seats, quantity, delivery type), 194 venues with capacity and coordinates, and 265 performers**, with no personal data. Prices are locked in the free sample, so prices will be set per tier (VIP, Premium, Silver, Bronze) based on real Kigali tiers. Customers, bookings and gate scans will be generated as test data on top of the real catalogue.

4. Planned Tables

Table	Purpose / key columns
VENUES, SECTIONS, PRICE_TIERS	Venue, city, capacity, lat/long; sections per venue; each section belongs to a tier
EVENTS, PERFORMERS, EVENT_PERFORMERS	Event, category, start/end, status, sales window (FK venue); M:N bridge to performers
EVENT_PRICES	Price in RWF per event and tier
TICKETS	One row per seat per event; status AVAILABLE / HELD / SOLD / USED / CANCELLED; UNIQUE (event, section, row, seat)
CUSTOMERS, BOOKINGS, BOOKING_ITEMS	Anonymous buyer; booking header and total; one ticket per item (UNIQUE ticket_id)
PAYMENTS, GATE_SCANS	MoMo / card / cash payment per booking; every gate scan with result ADMITTED / DENIED
PUBLIC_HOLIDAYS, AUDIT_LOG	Dates for the restriction rule; user, action, table, time, ALLOWED / DENIED, message

5. Planned PL/SQL Features

Feature	Where it is used
Procedures	sp_book_tickets puts seats on hold; sp_record_payment confirms the booking only if the amount equals the total; sp_cancel_booking releases the seats; sp_scan_ticket admits a ticket once
Functions	fn_available_seats(event, tier); fn_booking_total(booking); fn_occupancy_pct(event); fn_customer_ticket_count(customer, event)
Cursors	An explicit cursor in sp_release_expired_holds cancels unpaid holds older than 15 minutes; BULK COLLECT builds the per-event sales report
Packages	pkg_ticketing (booking, payment, exceptions), pkg_gate (entry), pkg_reports (RANK of events by revenue, LAG for daily sales change)
Triggers	trg_no_double_sale blocks selling a seat that is already sold; trg_ticket_limit (compound) blocks more than 6 tickets per customer per event; trg_event_open blocks booking after the event starts; trg_single_entry rejects a second scan; trg_restrict_dml blocks DML on weekdays and holidays; audit triggers log every action
Exceptions	-20001 seat unavailable, -20002 limit exceeded, -20003 event closed, -20004 already scanned, -20005 payment mismatch, -20010 restricted day; denied attempts are logged through an autonomous transaction

6. Innovation

Option: Analytics dashboard. Tool: Power BI Desktop, connected live to the Oracle PDB in DirectQuery mode (Microsoft, 2026). It has three pages: **Sales** (revenue, tickets by tier and category, daily trend), **Occupancy** (sold vs capacity, a venue map, events close to selling out) and **Gate & Security** (admitted vs denied scans, blocked scalping and double-sale attempts from AUDIT_LOG). Power BI only reads data; the PL/SQL enforces the rules.

7. References (APA)

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